

Established

- 12 COMSOL geometries pass matrix checks
- Multi-window + Kron + CUDA loop runs
- Physics inversion reaches ~1% response error

Promising, not proven

- GNN lowers nominal response error
- Fast direct estimate / useful initializer
- Turn-level graph accepts C and L/M edges

Still missing

- No hardware DAB validation
- No multi-seed uncertainty
- GNN does not beat physics accuracy